

Registration No.: _____

IMPORTANT NOTE

- 1) This sample paper serves solely as a visual representation of the question paper.
- 2) Each question in the question paper is mapped with corresponding Course Outcome(s) (CO) and Revised Bloom's Taxonomy (RBT) Level. However the question-wise CO-RBT mapping given below is indicative only and will vary in the final question paper.
- 3) Students should refer to the latest course syllabus and question paper pattern for exam preparation.
- 4) The final version of the End Term Examination question paper may differ in terms of layout, design, and content. This sample paper should be viewed for reference purpose only and not as final/official question paper.

24252INT58147

Paper Code: A

Course Code: INT428

Course Title: ARTIFICIAL INTELLIGENCE ESSENTIALS

Time Allowed: 03:00hrs.

Max Marks: 60

Read the following instructions carefully before attempting the question paper.

1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 60 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. Attempt all the questions in serial order.
4. Do not write or mark anything on the question paper and/or on rough sheet(s) which could be helpful to any student in copying, except your registration number on the designated space.
5. Submit the question paper and the rough sheet(s) along with the OMR sheet to the invigilator before leaving the examination hall.

Q(1) Locate the major difference between Bayesian networks and rule-based systems.

- (a) Rule-based systems handle probabilistic reasoning better than Bayesian networks.
- (b) Bayesian networks incorporate uncertainty, while rule-based systems rely on deterministic logic.
- (c) Bayesian networks cannot be used in AI applications.
- (d) Rule-based systems are the only approach for AI reasoning.

CO3,L2

Q(2) Select the best strategy for refining AI-generated responses.

- (a) Rewriting and iterating prompts
- (b) Using a single-word input
- (c) Asking for random outputs
- (d) Limiting AI to a single response type

CO5,L2

Q(3) What is a production system?

- (a) A system of rules for problem-solving
- (b) A hardware-based solution framework
- (c) A system for managing production tasks
- (d) A framework for data storage

CO1,L1

Q(4) Certainty factors are used in:

- (a) Rule-based expert system
- (b) Bayesian neural networks
- (c) Genetic algorithms
- (d) Deep reinforcement learning

CO3,L2

Q(5) Select the correct application of Natural Language Processing

- (a) Self-driving cars
- (b) Image recognition
- (c) Machine translation
- (d) Circuit design

CO4,L2

Q(6) Classify the following into characteristics of human intelligence and artificial intelligence

- (a) Problem-solving, decision-making
- (b) Emotional responses, self-awareness
- (c) Data processing, pattern recognition
- (d) Both (2) and (3)

CO1,L2

Q(7) Identify the component of a GAN that is responsible for generating new samples.

- (a) Encoder
- (b) Discriminator
- (c) Decoder
- (d) Generator

CO5,L1

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Q(8) What is a key characteristic of an "alpha" value in Alpha-beta pruning?

- (a) It represents the best possible score for the maximizer along the current path.
- (b) It represents the best possible score for the minimizer along the current path.
- (c) It is a fixed score used to break ties between moves.
- (d) It is a random value used for pruning decisions.

CO4,L2

Q(9) What does a Bayesian network represent?

- (a) Probabilistic relationships among variables
- (b) Deterministic rules in logic
- (c) Knowledge graphs in AI
- (d) Fuzzy logic statements

CO3,L2

Q(10) What is the maximum number of possible states in a water jug problem with two jugs of sizes 4 liters and 3 liters?

- (a) 7
- (b) 12
- (c) 15
- (d) 20

CO1,L2

Q(11) In AI, the "branching factor" in state-space search refers to:

- (a) The number of paths leading to the goal state
- (b) The number of heuristic values used
- (c) The number of different states generated from a given state
- (d) The depth of the search tree

CO1,L2

Q(12) Determine the correct visualization method for categorical data.

- (a) Line chart.
- (b) Scatter plot.
- (c) Bar chart.
- (d) Heatmap.

CO6,L2

Q(13) Why is the 8-puzzle problem considered a search problem in AI?

- (a) It requires intelligent agents to understand human language
- (b) It involves systematic exploration of a state-space to reach a goal state
- (c) It can only be solved using machine learning algorithms
- (d) It does not involve any form of search strategy

CO1,L1

Q(14) Tell the main difference between Greedy Best-First Search and A* Search.

- (a) A* considers both cost and heuristic, GBFS only heuristic.
- (b) GBFS finds the optimal path.
- (c) A* is faster than GBFS.
- (d) GBFS uses cost but ignores heuristics.

CO2,L1

Q(15) Which error occurs when dividing by zero?

- (a) ValueError
- (b) ZeroDivisionError
- (c) NameError
- (d) IndexError

CO6,L1

Q(16) What assumption is typically made in Naive Bayes classification?

- (a) All features are correlated.
- (b) Equal priors for all classes.
- (c) No missing values.
- (d) All features are independent given the class.

CO3,L2

Q(17) Given two jugs (Here L denotes litres):

J1 with a maximum capacity of 9L

J2 with a maximum capacity of 5L

The current state of water in the jugs is $(X, Y) = (6, 2)$, where:

$X = 6$ (J1 contains 6L)

$Y = 2$ (J2 contains 2L)

If we apply the following two operations in sequence:

1. Fill J2 to its maximum capacity
2. Pour water from J2 to J1

Identify the next state

- (a) (8,5)
- (b) (9,2)
- (c) (6,0)
- (d) (7,5)

CO1,L2

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Q(18) Which of the following is a key technique for error identification when using ChatGPT for data analysis tasks?

- (a) Using data validation methods to identify discrepancies or unusual patterns in input data
- (b) Relying on human intuition to guess errors
- (c) Ignoring errors if the data set is too large
- (d) Only checking for formatting issues in the text output

C06,L2

Q(19) Select the key difference between Generative AI and traditional AI models.

- (a) Generative AI is rule-based, while traditional AI is data-driven
- (b) Generative AI creates new data, while traditional AI analyzes existing data
- (c) Generative AI is slower than traditional AI
- (d) Generative AI does not require machine learning

C05,L2

Q(20) Best-First Search selects the next node to expand based on:

- (a) The shortest path found so far
- (b) The heuristic value of the node
- (c) The number of nodes visited
- (d) A random selection

C02,L2

Q(21) Show how to solve the issue of duplicate rows in a dataset. Which method is correct?

- (a) df.fillna()
- (b) df.head()
- (c) df.count()
- (d) df.drop_duplicates()

C06,L2

Q(22) Siri, Alexa, and Cortana are examples of:

- (a) Machine learning algorithms
- (b) Personal voice assistants
- (c) Web search engines
- (d) Data compression techniques

C04,L2

Q(23) Choose the most suitable AI-based writing assistant.

- (a) Canva
- (b) Grammarly
- (c) Photoshop
- (d) Audacity

C06,L2

Q(24) In a Minimax algorithm, what does the MAX player try to do?

- (a) Minimize the evaluation function
- (b) Maximize the evaluation function
- (c) Maximize the number of moves
- (d) Minimize the depth of the search tree

C04,L1

Q(25) Choose the odd one out with respect to knowledge representation?

- (a) Heuristic Search
- (b) Semantic Network
- (c) Frames
- (d) Logical Representation

C03,L1

Q(26) Which of the following is an approach in knowledge representation in AI?

- (a) Frames
- (b) Neural networks
- (c) Decision trees
- (d) Genetic algorithms

C03,L2

Q(27) Which of the following is NOT a foundation of AI?

- (a) Mathematics
- (b) Psychology
- (c) Philosophy
- (d) Astrology

C01,L2

Q(28) What would be the Soundex code for the name "Smith" based on the Soundex rules?

- (a) S300
- (b) S530
- (c) S123
- (d) S100

C04,L2

Q(29) What is an informed search strategy?

- (a) A search that does not require any additional knowledge
- (b) A search that uses heuristics to guide the search process
- (c) A search that expands all nodes equally
- (d) A search that always follows a predefined path

C02,L2

Q(30) What is a prompt in AI?

- (a) A type of algorithm
- (b) An input given to a language model
- (c) A response from a chatbot
- (d) A deep learning model

C05,L2

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Q(31) Outline the sequence of NLP phases in chatbot conversation processing.

- (a) Response generation → Intent recognition → Text preprocessing → Speech recognition
- (b) Intent recognition → Speech recognition → Response generation → Text preprocessing
- (c) Speech recognition → Text preprocessing → Intent recognition → Response generation
- (d) Text preprocessing → Response generation → Intent recognition → Speech recognition

CO4,L2

Q(32) Select the most appropriate option that best defines production systems for “Chemical Synthesis”:

- (a) Monotonic and Partially Commutative
- (b) Nonmonotonic and Partially Commutative
- (c) Nonmonotonic and Not Partially Commutative
- (d) Monotonic and Not Partially Commutative

CO1,L2

Q(33) Given an 8-puzzle, if the Manhattan distance heuristic $h(n)$ at a state is 5 and the cost $g(n)$ is 3, what is $f(n)$ in A search?*

- (a) 8
- (b) 5
- (c) 3
- (d) 24

CO2,L2

Q(34) Which of the following statements demonstrates De Morgan’s Law?

- (a) $\neg(P \vee Q) = (\neg P) \wedge (\neg Q)$
- (b) $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$
- (c) $P \rightarrow Q$ is equivalent to $\neg P \vee Q$
- (d) $P \leftrightarrow Q$ is equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow P)$

CO3,L2

Q(35) What approach would you use to process media files in ChatGPT Advanced Data Analysis?

- (a) Extract text and metadata using AI models
- (b) Convert media to text manually
- (c) Compare different media formats
- (d) Avoid using media files

CO6,L2

Q(36) Which model is commonly used for text generation?

- (a) RNN
- (b) LSTM
- (c) Transformer
- (d) All of the above

CO5,L2

Q(37) What is the main purpose of Alpha-Beta pruning?

- (a) To improve heuristic evaluation
- (b) To reduce the number of nodes evaluated in Minimax
- (c) To randomly select the best move
- (d) To ensure fairness in game playing

CO4,L2

Q(38) How would you use NLP techniques to improve virtual assistants like Alexa or Siri?

- (a) Implement Minimax search for decision-making.
- (b) Use NLP techniques like Named Entity Recognition (NER) and Sentiment Analysis.
- (c) Apply Dijkstra’s Algorithm to process speech.
- (d) Use Alpha-Beta Pruning to reduce computation time.

CO4,L2

Q(39) Identify the significance of $h^*(n)$ in the given equation $h(n) \leq h^*(n)$

- (a) Heuristic estimate of the cost from node n to the goal node
- (b) Actual cost from node n to the goal node
- (c) Heuristic estimate of the goal node
- (d) Heuristic estimate of the starting node

CO2,L2

Q(40) Why does Simulated Annealing accept worse solutions with a certain probability during the early stages?

- (a) To ensure the algorithm runs faster
- (b) To escape local minima and explore more of the solution space
- (c) To always move towards the global minimum
- (d) To avoid using cooling schedules

CO2,L2

Q(41) What is the primary purpose of Variational Autoencoders (VAEs)?

- (a) To classify images into predefined categories
- (b) To generate new data by learning a probabilistic latent space
- (c) To detect anomalies in time-series data
- (d) To optimize database queries

CO5,L1

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Q(42) Determining the advantage of backward chaining.

- (a) It uses probabilistic reasoning.
- (b) It is goal-driven.
- (c) It stores knowledge in a network.
- (d) It is purely statistical.

C03,L2

Q(43) How do GPT models differ from search engines?

- (a) They generate new text
- (b) They retrieve live results
- (c) They rank search links
- (d) They collect websites

C05,L2

Q(44) Which of the following search algorithms guarantees the optimal solution, assuming the heuristic is admissible?

- (a) A* Algorithm
- (b) Simulated Annealing
- (c) Hill Climbing
- (d) Best-First Search

C02,L2

Q(45) How does Generative AI "learn" to create new content?

- (a) By memorizing past data
- (b) By detecting patterns in training data
- (c) By storing all previous outputs
- (d) By manually coding responses

C05,L2

Q(46) How would you compare PaLM with smaller language models?

- (a) PaLM cannot be fine-tuned for specific applications
- (b) Smaller language models always perform better than PaLM
- (c) PaLM has more parameters, which allows it to understand complex queries better
- (d) Smaller models require more data to function effectively

C05,L2

Q(47) In Dempster-Shafer Theory, belief functions represent:

- (a) The degree of confidence in a hypothesis
- (b) The number of possible outcomes
- (c) A fixed probability of an event occurring
- (d) A deterministic outcome with no uncertainty

C03,L2

Q(48) Name the approach that is commonly used for spell checking in NLP.

- (a) Edit Distance
- (b) TF-IDF
- (c) Tokenization
- (d) Latent Dirichlet Allocation

C04,L2

Q(49) How does Random Restart Hill Climbing overcome the issue of getting stuck at local maxima?

- (a) By restarting the search from multiple random initial states
- (b) By making only small changes in the search path
- (c) By avoiding heuristic evaluation
- (d) By selecting only one best path at the start

C02,L2

Q(50) Which of the following best describes the difference between ChatGPT and ChatGPT ADA?

- (a) ADA is used only for writing fiction
- (b) ADA provides enhanced visual rendering
- (c) ADA can analyze datasets, while standard ChatGPT cannot
- (d) Both can run Python code equally

C06,L2

Q(51) Choose the odd one out among the following categories of machine learning.

- (a) Unsupervised Learning
- (b) Semi-Supervised Learning
- (c) Reinforcement Learning
- (d) Static Learning

C06,L1

Q(52) When is Depth-First Search preferred over Breadth-First Search?

- (a) When graphs have uniform costs.
- (b) When memory constraints are strict.
- (c) When optimal solutions are needed.
- (d) When paths have equal lengths.

C02,L2

Q(53) Differentiate between the Minimax algorithm and Alpha-Beta pruning in AI-based game playing.

- (a) Minimax evaluates all possible game states, while Alpha-Beta pruning skips unnecessary branches.
- (b) Alpha-Beta pruning is an alternative to Minimax that replaces evaluation functions.
- (c) Minimax only works with games like Chess, while Alpha-Beta pruning is for real-time strategy games.
- (d) Alpha-Beta pruning eliminates the need for Minimax in AI game-playing.

C04,L2

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Q(54) Select the option that does NOT represent a valid logical connector in propositional logic.

- (a) Conjunction (b) Disjunction (c) Aggregation (d) Implication

CO3,L2

Q(55) What would result if error-handling techniques were not applied when using ChatGPT Advanced Data Analysis for data processing?

- (a) The system would provide clear and structured results
(b) The process would become more efficient
(c) The AI would automatically fix all errors without intervention
(d) The output might include incorrect or incomplete data

CO6,L2

Q(56) Which search algorithm is suitable for problems with infinite state spaces?

- (a) BFS (b) DFS
(c) Iterative Deepening (d) Bi-Directional Search

CO2,L2

Q(57) Which of the following best defines intelligence?

- (a) The ability to perform a task without learning
(b) The ability to adapt to new situations and solve problems
(c) The ability to memorize information
(d) The ability to follow instructions precisely

CO1,L2

Q(58) Which of these is a prompt pattern?

- (a) Instruction-based prompt (b) Chain-of-thought prompt
(c) Few-shot prompt (d) All of the above

CO5,L2

Q(59) Determine the most suitable AI tool for AI in Marketing and Social Media Content Creation.

- (a) Photoshop (b) Audacity (c) Excel (d) ChatGPT

CO6,L2

Q(60) Which of the following best describes a problem in AI?

- (a) A rule that prevents any state changes (b) A task defined by an initial and goal state
(c) A process that avoids exploring states (d) A goal that does not require searching

CO1,L2

--End of Question paper--